Review: Operations with rational numbers

1 Evaluate the following:

a $-7+9.1$

b $4-6.1$

c $-13+45.31$

d $29-36.71$

e -6×0.8

f -13.8×9

g $-2+1.9+7.3$

h $-6.8-(-1.1)+8.8$

i $2.8+6\times(-9.5)$

j $24.7-9.9-8.2\div(-2)$

k $\dfrac{9}{-36}(9+(-6))$

l $86+\dfrac{3}{10}\times(-2)$

2 Is the product of two irrational numbers always irrational?

3 David buys 3 shirts at $\$19.90$ each, and a pair of jeans for $\$20.50$. The shop has a sale on, and so he receives a $\$8.02$ discount. Find the total amount he spends.

4 After an earthquake, $\dfrac{4}{7}$ of all claims were paid within a month. Of the remaining claims not paid $\dfrac{4}{5}$ of them were paid within the second month. Calculate the fraction of all claims paid in the second month.

5 At the end of last month, a dam was $\dfrac{4}{7}$ full. Over the current month, $\dfrac{37}{56}$ of the dam's capacity was used up by the town. Meanwhile, rain filled $\dfrac{5}{8}$ of the dam's capacity over the duration of the month. Calculate the fraction of the dam filled at the end of this month.

Additional questions

6 Simplify the following fractions:

a $\dfrac{3}{9}$

b $\dfrac{20}{60}$

c $\dfrac{6}{20}$

d $\dfrac{27}{45}$

e $\dfrac{63}{28}$

f $\dfrac{40}{30}$

g $\dfrac{26}{50}$

h $\dfrac{24}{66}$

i $\dfrac{32}{80}$

j $\dfrac{64}{92}$

k $\dfrac{80}{100}$

l $\dfrac{60}{200}$

m $\dfrac{2}{10}$

n $\dfrac{8}{64}$

o $\dfrac{21}{231}$

p $\dfrac{150}{270}$

7 Simplify the following fractions, writing your answers as mixed numbers:

a $\frac{11}{7}$ b $\frac{24}{10}$ c $\frac{18}{4}$ d $\frac{63}{45}$

8 Evaluate the following, writing your answers in simplest form:

a $\frac{5}{9}+\frac{8}{9}$ b $\frac{1}{5}+\frac{2}{4}$ c $\frac{7}{8}+\frac{5}{6}$ d $\frac{6}{5}+\frac{5}{3}$

e $\frac{1}{3}-\frac{1}{10}$ f $\frac{3}{4}+\frac{1}{8}-\frac{1}{4}$ g $\frac{5}{9}+\frac{2}{3}+\frac{5}{6}$ h $\frac{8}{9}\times\frac{9}{56}$

i $\frac{8}{15}\times\frac{9}{7}$ j $\frac{5}{3}\times\frac{21}{2}$ k $\frac{9}{10}\times\frac{10}{9}$ l $\frac{11}{9}\times\frac{2}{99}$

m $-\frac{3}{7}\times\frac{2}{3}$ n $\left(\frac{4}{9}\right)^2$ o $\frac{2}{7}\div\frac{5}{3}$ p $\frac{9}{10}\div\frac{10}{9}$

q $\frac{2}{7}\div\frac{7}{18}$ r $\frac{3}{15}\div\frac{4}{3}$ s $\frac{3}{8}\div\frac{3}{4}$ t $\frac{11}{35}\div\frac{55}{49}$

u $\left(-\frac{4}{5}\right)\div\frac{3}{4}$ v $\frac{3}{5}\times\frac{2}{3}+\frac{5}{15}$ w $\left(\frac{2}{3}+\frac{3}{5}\right)\times\frac{3}{7}$

9 Evaluate the following, writing your answers as mixed numbers in simplest form where necessary:

a $2\frac{3}{11}+4\frac{7}{11}$ b $3\frac{2}{6}+4\frac{4}{5}$ c $3\frac{1}{3}+4\frac{11}{12}$ d $2\frac{2}{3}+3\frac{11}{15}$

e $6+1\frac{4}{5}$ f $7+3\frac{3}{4}$ g $11+4\frac{3}{8}$ h $4-\frac{7}{9}$

i $5\frac{10}{11}-4\frac{8}{11}$ j $4\frac{6}{7}-3\frac{6}{8}$ k $6\frac{1}{2}-4\frac{1}{10}$ l $6\frac{1}{2}-\frac{3}{8}$

m $2-\frac{3}{4}$ n $8-2\frac{1}{2}$ o $10-5\frac{1}{3}$ p $9-2\frac{5}{7}$

q $1\frac{1}{2}\times\frac{2}{3}$ r $1\frac{3}{5}\times2\frac{1}{4}$ s $1\frac{3}{4}\times\frac{4}{9}$ t $2\frac{1}{4}\times3\frac{1}{3}$

u $1\frac{2}{5}\div\frac{3}{4}$ v $2\frac{1}{4}\div1\frac{2}{5}$ w $1\frac{2}{7}\div3\frac{1}{2}$ x $3\frac{1}{3}\div2\frac{1}{2}$

10 Evaluate:

a $\left(\frac{4}{7}+\frac{2}{5}\right)\div\frac{5}{8}$

b $\frac{4}{16}-\frac{2}{16}+\frac{12}{16}$

c $\frac{4}{35}-\left(\frac{6}{7}-\frac{4}{5}\right)$

d $\left(\frac{2}{5}-\frac{3}{8}\right)\times\frac{5}{9}$

e $\frac{47}{72}-\left(\frac{2}{8}+\frac{3}{9}\right)$

f $\frac{6}{7}\times\left(\frac{2}{3}-\frac{2}{7}\right)$

g $\frac{7}{8}\div\left(\frac{3}{8}+\frac{2}{7}\right)$

h $\frac{5}{8}\div\left(\frac{6}{7}-\frac{2}{9}\right)$

i $\left(\frac{3}{5}-\frac{1}{4}\right)\div\frac{1}{2}$

j $\left(\frac{1}{4}+\frac{3}{8}\right)\times\left(\frac{2}{5}+\frac{1}{5}\right)$

k $\frac{5}{7}\times\left(\frac{1}{3}+\frac{5}{6}-\frac{1}{2}\right)$

l $\left(\frac{2}{5}+\frac{3}{10}\times\frac{2}{3}\right)\div\frac{6}{7}$

m $\left(\frac{1}{3}+\frac{3}{4}\right)\div\left(\frac{1}{2}-\frac{3}{8}\right)$

n $8\frac{1}{4}-2\frac{1}{5}\times3\frac{1}{2}$

11 Evaluate:

a $46.01+0.09$

b $7.411+6.247$

c $12.89+34.11$

d $36.97-21.62$

e $8.79-6.33$

f 0.5×1000

g 0.471×5

h 2.2×4

i 1.2×3.2

j 4.23×2

k 4.8×8

l 2.68×2.31

m $\frac{0.005\,34}{0.006}$

n $\frac{0.247}{0.019}$

o $9.2768\div2.08$

p $56\div0.7$

q $1.2\div0.3$

r $420\div0.7$

s $8.1\div0.9$

t $393.68\div7$

u $2.270\,46\div0.474$

v $36.53\times63\div9$

w $6.42-8.2\times0.7+4.67$

12 Round the following to the nearest cent:

a $\$34.7691$

b $\$64.4725$

c $\$254.899$

d $\$32.455$

13 Round the following to the nearest five cents.

a \$27.82	**b** \$64.86	**c** \$34.89	**d** \$57.54
e \$227.02	**f** \$634.67	**g** \$1024.98	**h** \$21.94

14 For the following expressions:

i Estimate the answer by rounding both values to the nearest whole number.

ii Find the answer using a calculator.

iii Hence, find the difference between the estimate and the actual answer.

iv Suggest a way to improve the accuracy of your estimate.

a $6.61 + 2.85$ b $9.66 - 7.88$ c $11.59 \div 4.44$

15 For each of the following expressions:

i Estimate the answer by rounding both values to the nearest whole number.

ii Find the answer using a calculator. Round your answer to two decimal places.

a 1.29×3.42 b $26.99 \div 8.98$

16 Dave used a calculator to evaluate 6.573×8.03 but forgot to enter the decimal points. The answer he got was $5\,278\,119$. What should the answer have been?

17 Pauline is making 5 chocolate coated strawberries for her friends. She has 52.5 g of chocolate which she uses to evenly coat the strawberries.

a One of the strawberries has a mass of 19 g. Choose the most reasonable estimate for its mass once it is coated in chocolate:

A 29 g **B** 69 g **C** 9 g **D** 10 g

b Calculate the exact mass of the 19 g strawberry after it is coated in chocolate.

18 William earns $\$304.92$ for 5 days of work.

a Choose the most reasonable estimate for how much William earns per day:

A $\$6.10$ **B** $\$610.00$ **C** $\$61.00$ **D** $\$152.50$

b Calculate the exact amount William earns each day.

19 Katrina and Luigi ordered a pizza to share. Katrina ate $\frac{3}{7}$ of the pizza while Luigi ate $\frac{1}{2}$.

a Who ate more pizza?

b What fraction of the pizza did they eat together?

c What fraction of pizza is left?

20 Han spent $\frac{1}{8}$ of his money on a printer and $\frac{1}{5}$ of his money on a table.

a What fraction of his total money did Han spend?

b What fraction of Han's money does he still have?

21 A class has 40 students. $\frac{2}{5}$ of the class are boys.

a What fraction of the class is girls?

b How many girls are in the class?

22 Luke has a dice tin that is $\frac{1}{4}$ full. He puts 8 extra dice in and the tin is now $\frac{1}{3}$ full.

a What fraction of the tin do the 8 extra dice take up?

b How many dice can fit inside the tin?

23 In a survey of 270 people, $\frac{1}{10}$ said their favourite sport was soccer, and $\frac{1}{9}$ said their favourite sport was tennis.

a What is the total fraction of people who said their favourite sport was soccer or tennis?

b What is the fraction of people who did not put soccer or tennis as their favourite sport?

c How many people did not put soccer or tennis as their favourite sport?

24 Victoria borrows a 360-page novel from the library. She reads $\frac{3}{4}$ of the novel.

a How many pages did she read?

b How many more pages does she have to read to finish her novel?

25 In the morning, a dam contains $635\ \mathrm{L}$ of wat[illegible]roughout the day, $\frac{1}{5}$ of the water is lost through evaporation and cattle drinking from it.

a How many litres of water are lost throughout the day?

b How many litres of water are left in the dam at the end of the day?

26 At the Easter Show, Tobias spends half an hour visiting chickens and $\frac{1}{3}$ of an hour visiting alpacas. What fraction of an hour has he spent visiting animals?

27 Sharon walks $\frac{2}{5}$ of a mile and runs $\frac{1}{7}$ of a mile to get to school. How many miles does she travel?

28 Carl has $\frac{3}{7}\ \mathrm{ft}$ of ribbon. After he used some ribbon for a present, he has $\frac{1}{4}\ \mathrm{ft}$ left. How much ribbon did he use on the present?

29 Amy is buying some supplies for the new school year. She buys an exercise book for $\$9.55$, a pack of pens for $\$6.12$ and a pencil case for $\$6.28$. How much does she spend in total?

30 Harry buys an item from the school canteen for $\$3.20$. If he pays for it with a five dollar note, how much change will he get back?

31 A customer gives the cashier $\$100$ to pay for a purchase of $\$64.95$. How much change should the cashier give the customer?

32 Brad is currently $\$18$ overdrawn in his account. He knows that he needs to pay a $\$61.61$ bill from this account tomorrow. How much does he need to add to this account so he can pay his bill tomorrow?

33 Sebastian owes $\$47.59$ to each of his 3 friends. How much money does he owe altogether?

34 Mae earns $\$36.43$ per hour working as a call center operator. If she works 20.5 hours per week, find her weekly wage.

35 If Skye earns $\$238.42$ for working 7 hours, how much does she earn per hour?

36 Lisa purchased a car and is paying monthly installments of $\$148.40$ to pay it off. If it takes her 60 months to pay it off, how much will she have spent on car repayments in total?

out of 4 customers order hot chocolate. If sh[redacted]s 4000 customers on a particular day, how many hot chocolates would she expect to sell?

38 Ellie wants to install a rectangular swimming pool in her backyard. If the space available for a pool in her backyard measures $10\frac{1}{4}$ m by 4 m, what will be the total area of the pool?

39 A bag of oranges weighing $\frac{8}{9}$ lb was divided among 6 children. Find the mass of the oranges given to each child.

40 Bianca has $\frac{5}{6}$ lb of beads. She shares the beads evenly between 3 friends. How many pounds of beads does each friend receive?

41 Three items weighing 3.41 lb, 2.58 lb and 5.79 lb are to be posted but are too heavy to send. By how much does the total weight exceed the 11.38 lb limit?

42 How many pieces of steel, each 4.25 m long, can be cut from a wire 153 m long?

43 A piece of cord 2.87 ft long is cut evenly into smaller pieces of 0.07 ft each. How many of these pieces can be cut?

44 How many 0.38 L bottles can be filled from a barrel which holds 41.8 L?

45 A bottle is $\frac{2}{7}$ full of cordial. If 230 mL of cordial is added to it, the bottle is $\frac{5}{6}$ full. How much cordial does the bottle hold when full?

46 During a sale, a number of computers were sold. 6 computers were sold at a loss, which represents $\frac{1}{6}$ of the total number of computers that were sold.
If 89 computers were up for sale, how many weren't sold?

47 How much would you pay for eight t-shirts if they cost $\$46.88$ each, and you receive a discount of $\$9.47$?

48 Sally is a sales assistant. She earns $\$192$ per week, plus a commission of $\frac{1}{3}$ on anything she sells. Last week, she sold $\$726$ of printers.

a Find Sally's commission for last week.

b Find Sally's pay for last week.

a How far will 30 calculators reach if laid end to end?

b Convert this length to meters.

50 Maria runs 50.2 m in 15 seconds. If she continues running at the same speed, how far can she run in 2 minutes?

51 Neville trains six days a week. Each weekday, he runs 2.4 km and on Saturday, he runs 4.6 km. How far does he run in a whole week?

52 Quentin recorded the following times in the 100 m running race over a whole season. How much quicker was his fastest time than his slowest time?

Race	Time (seconds)
1	10.59
2	11.03
3	10.45
4	10.38
5	10.63

53 Christa is creating a 3-minute slide show with pictures from her holiday. Each picture will be displayed for 4.5 seconds.

a How many seconds are in 3 minutes?

b How many pictures can she show in 3 minutes?

54 For a project, Ivan needs to cut a metal rod such that the longer piece is three times the length of the shorter piece.

a What fraction of the rod's total length will be made up of the shorter piece?

b If the rod is $\frac{3}{8}$ ft long, find the length of the shorter piece.

55 An engineer is estimating the length and width of a room using the length of his steps. The length of the room is $\frac{4}{5}$ of the width. He determined that the width is about 27 steps and that one of his steps is $\frac{2}{3}$ m long. Find the length of the room.

56 About $\frac{3}{10}$ of the cars in a city are less than 5 years old. About $\frac{1}{5}$ of the cars are between 5 and 10 years old. If there is a total of 1200 cars registered in the city, find the number of cars that are greater than 10 years old.

57 The mall parking lot has 800 parking spaces. $\frac{1}{4}$ of the parking spaces are for cars. On a particular day there were 200 cars and some trucks in the parking lot. The parking lot was $\frac{5}{8}$ full. How many trucks were in the parking lot?

58 At 10 pm, the temperature in Victoria is $62.1\degree \mathrm{F}$. Each hour the temperature decreases by $3.4\degree \mathrm{F}$. Find the temperature 5 hours later.

59 At midnight, the temperature in Darwin is $85.28\degree \mathrm{F}$. Each hour after that the temperature

60 A diver is 3.5 ft below the surface of the water. He descends 1.47 ft each second for 3 seconds, then rises $0.46 \frac{\text{ft}}{\text{s}}$ for 6 seconds. The level at the surface is 0 ft, and 1 ft below is represented by -1 ft.

a Find the depth of the diver after 9 seconds.

b The diver sees a turtle near the surface and needs to get to a depth of 0.35 ft below the surface within 4 seconds to snap the perfect shot. How many feet does he need to rise each second, assuming he will rise at a constant speed?